

Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 7



By- Ravindra Sir

Recap of Previous Lecture



Topic

Reg languages

Topic

Topic

Topics to be Covered



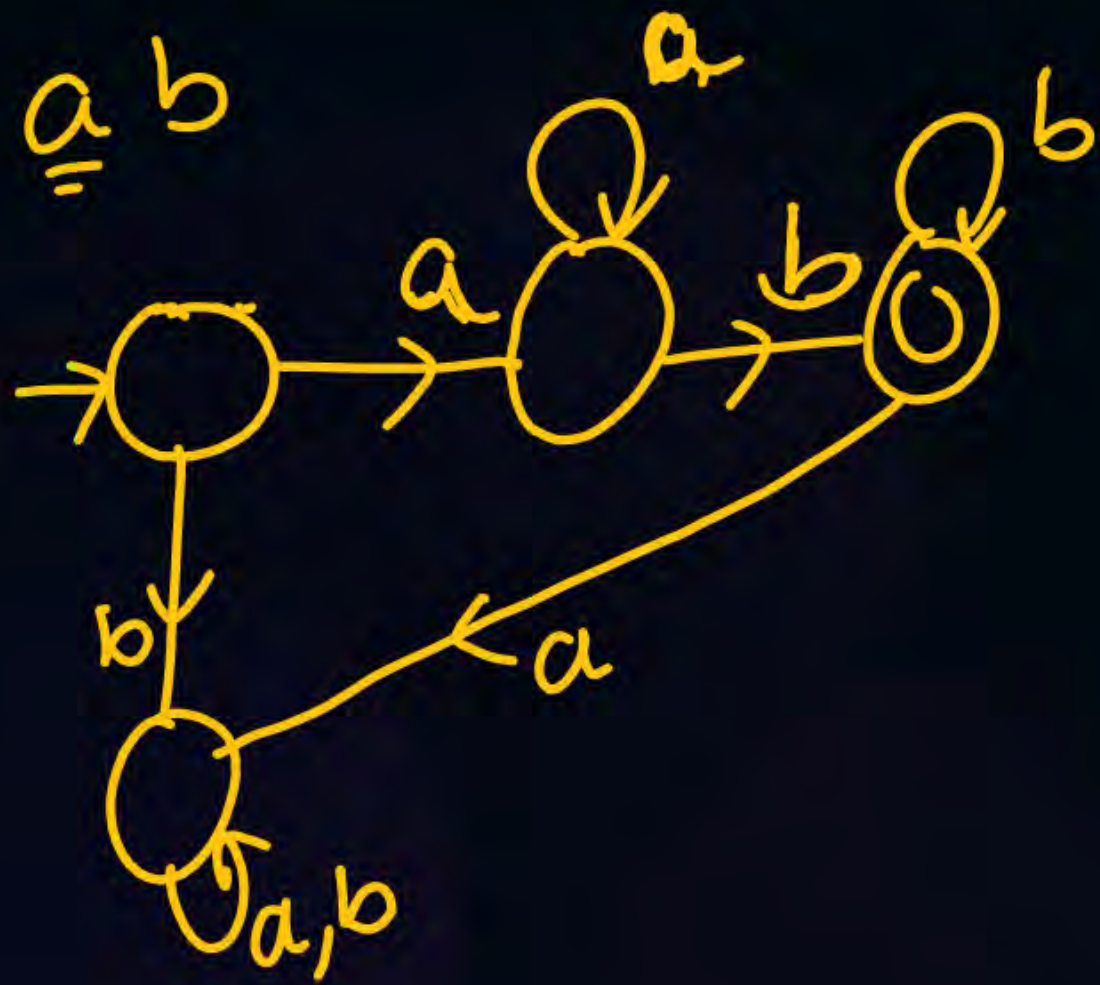
Topic

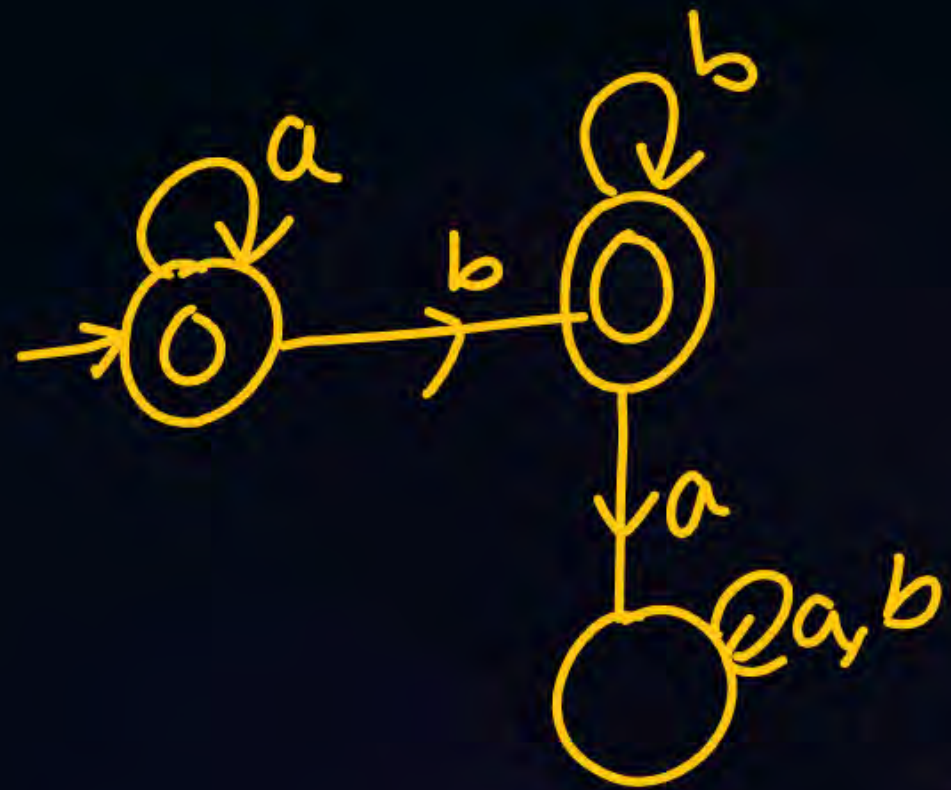
Regular languages

Topic

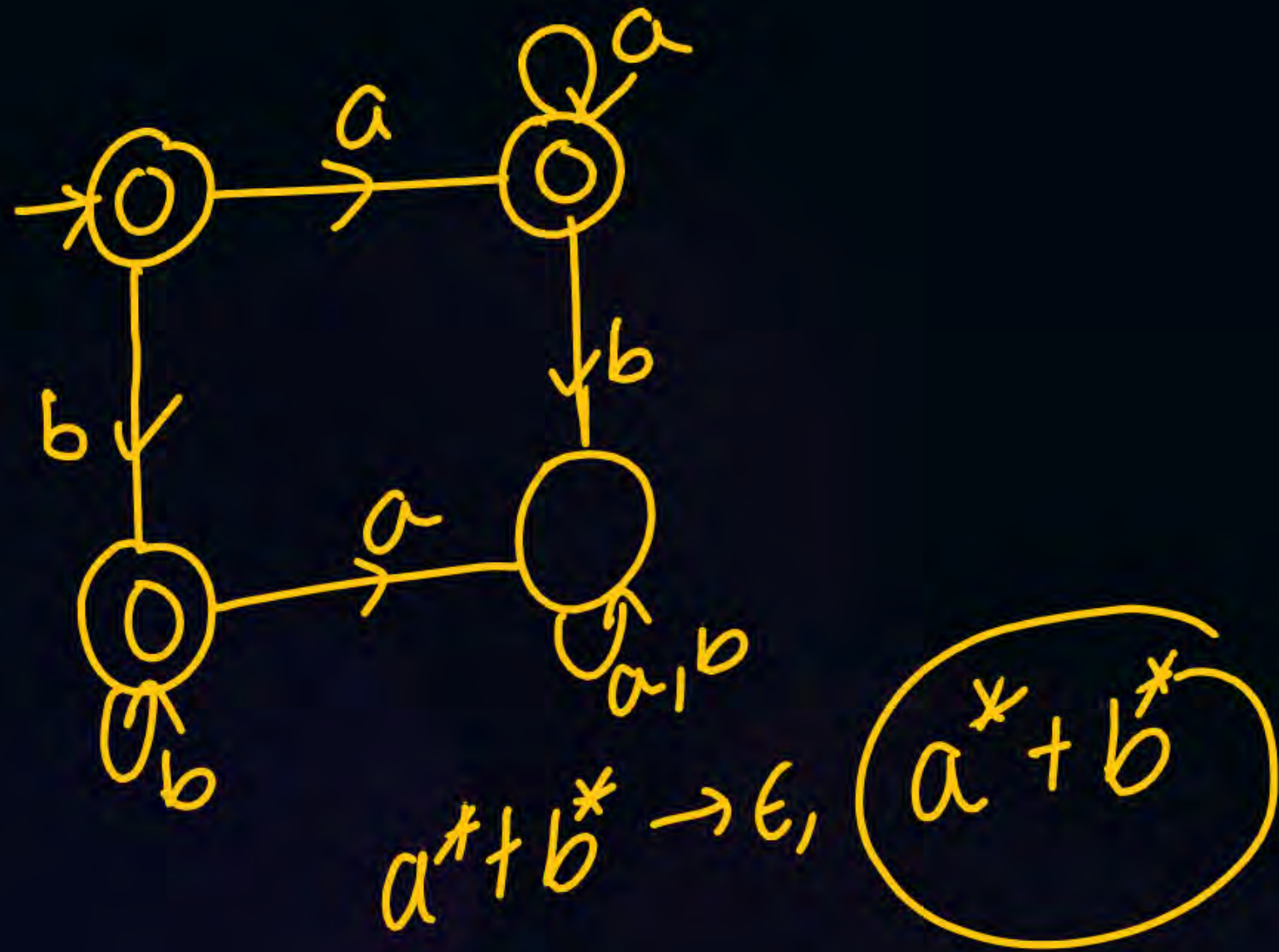
Topic

$$\underline{a^n b^m / n, m \geq 1}$$

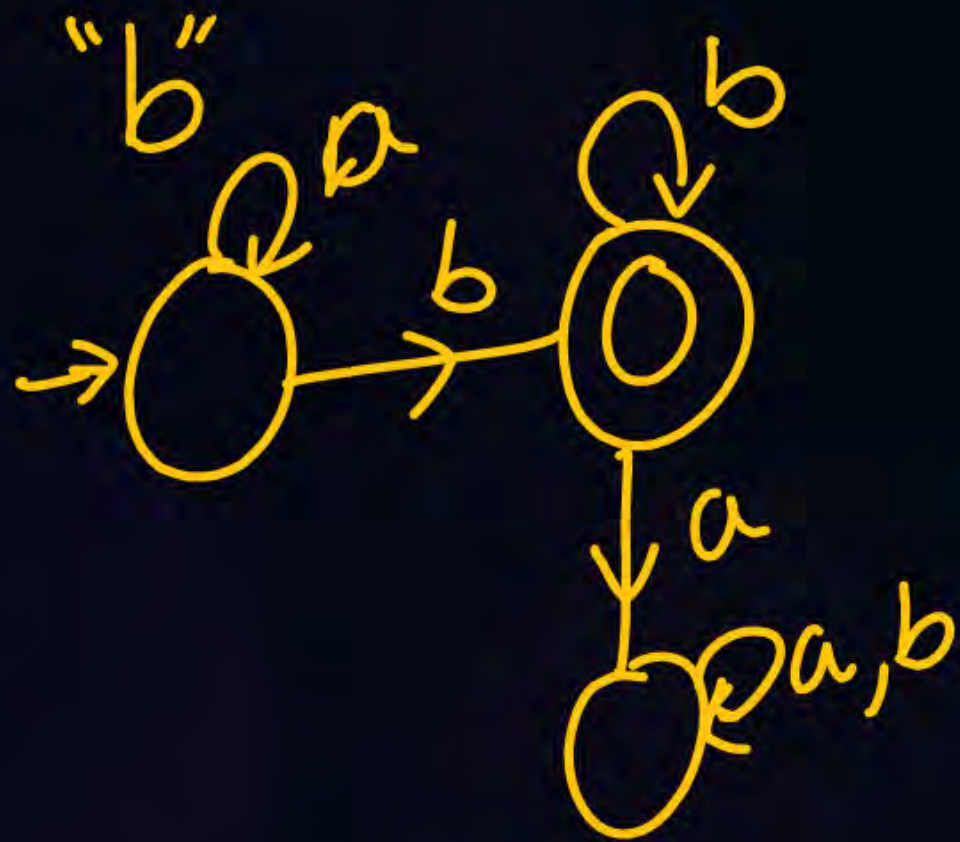


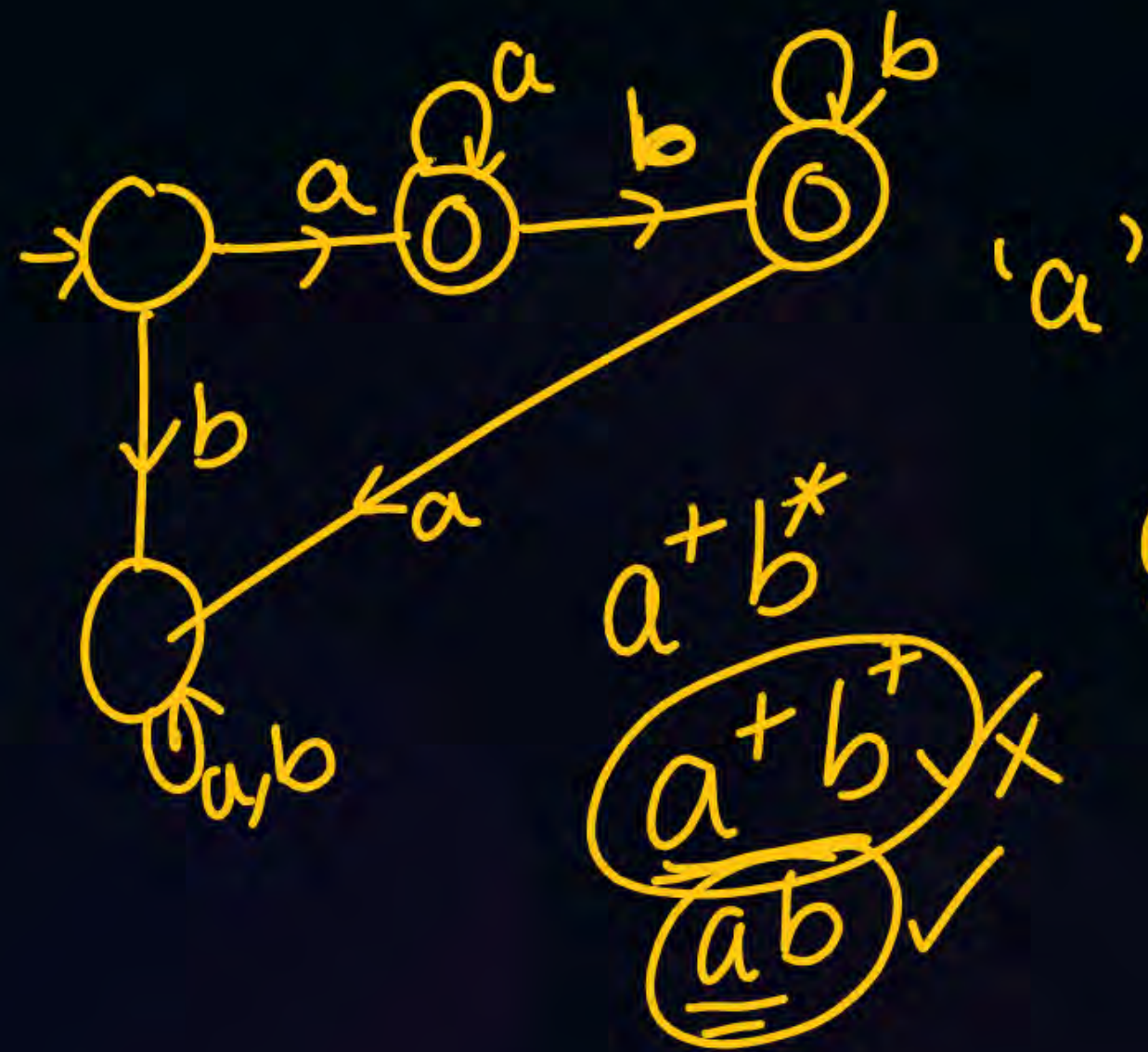


$$a^0 b^0 \rightarrow \underline{\underline{a^*, b^*}} \quad a^* b^*$$



$$a^n b^m / n \geq 0, m \geq 1$$





$$\frac{aa^* \rightarrow ob's}{a^+ \rightarrow ob's} \quad a^+b^*$$

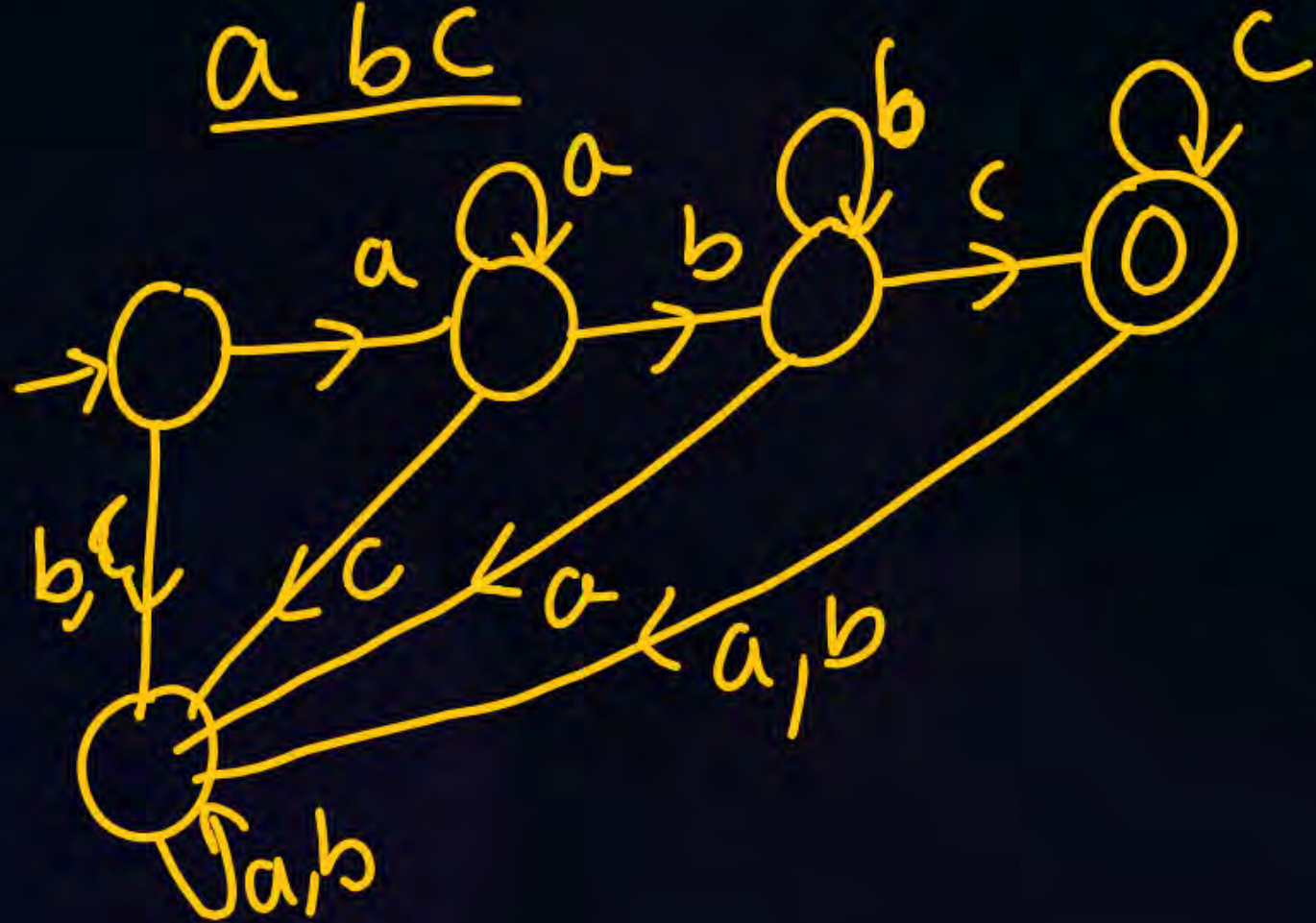
$$\frac{aa^*}{a} \quad \frac{aa^*b^*}{a^+b^*}$$

$$\frac{a^+b^*}{ab} \quad \checkmark$$

$$a^+b^+c^+$$

$$a^n b^m c^l / n, m, l \geq 1$$

abc



$$a^n b^m c^l / n, m, l \geq 0$$

$$a^* b^* c^*$$



✓
5 min

$$a^* b^* c^*$$

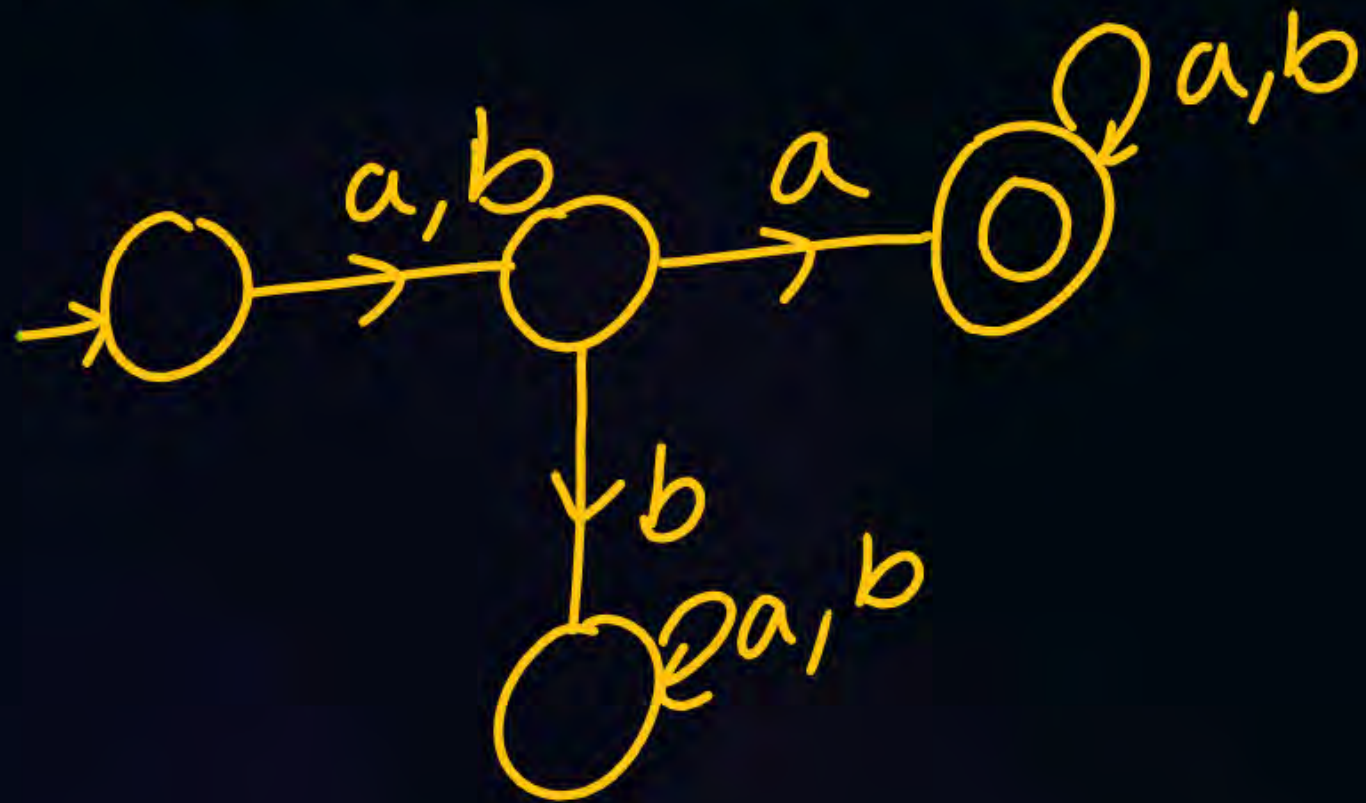
$$\rightarrow \underline{a}^*, \underline{b}^*, \underline{c}^* \checkmark$$

$$\Sigma = \{a, b\} \quad L = \{ \underline{a^3} b \omega \underline{a^3} / \omega \in (a, b)^* \}$$

5 min

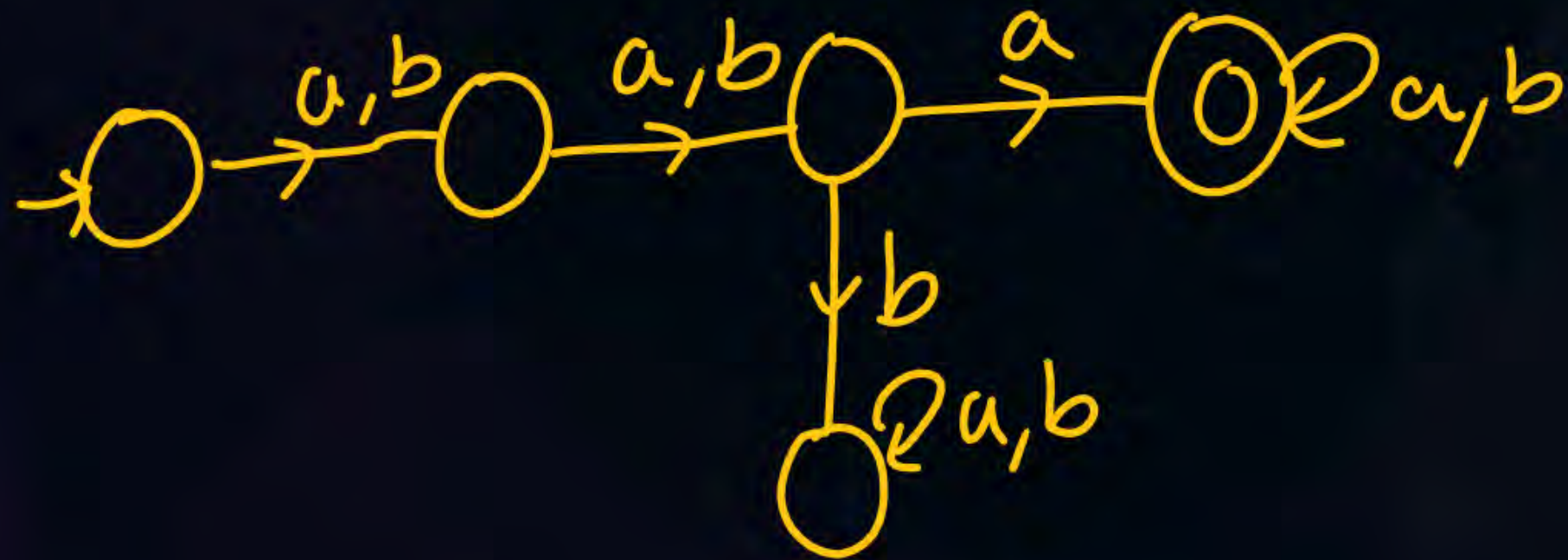


$L = \{w / \text{second symbol in } w \text{ is 'a'}\}$



$\Sigma = \{a, b\}$

III symbol is 'a' (III symbol from LHS is 'a')



NFA $\Rightarrow$ DFA ✓

$L = \{a, b\}$

Second symbol from RHS is 'a'

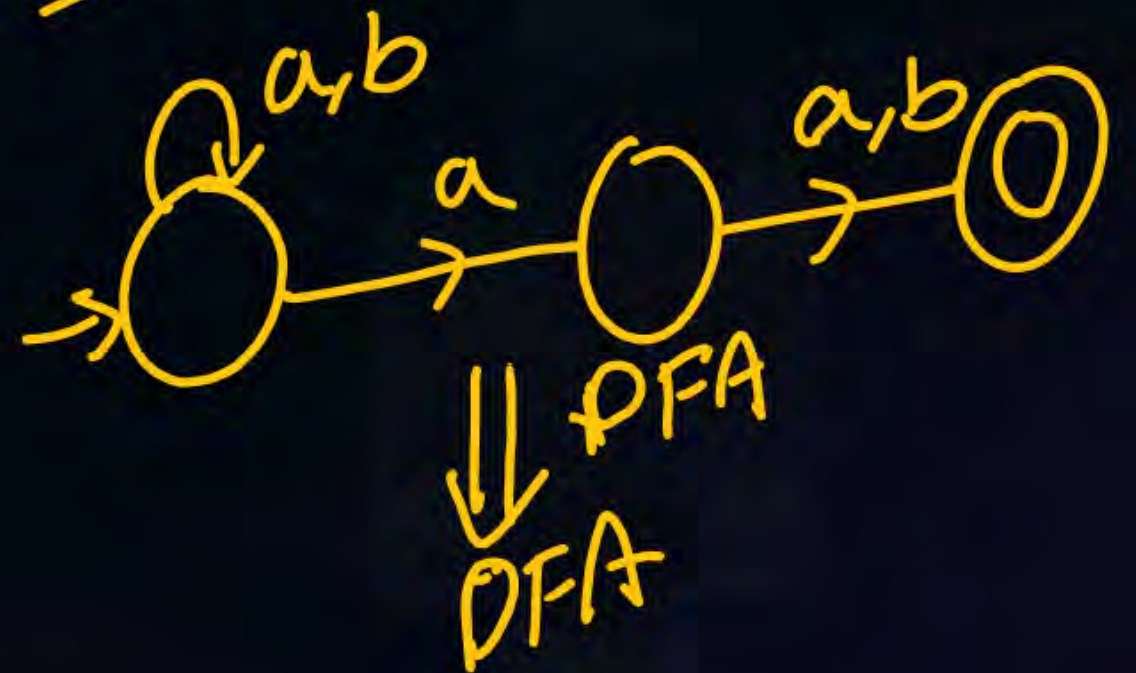
(3)

$(a+b)^* a (a+b)$

$(a+b)^* \underline{a} a + (a+b)^* \underline{a} \underline{b}$
ending with 'aa' or 'ab'

DFA - X

NFA:



✓
FA $\rightarrow$ RE:-

- (1) use intuition ✓
- (2) State elimination
- (3) Ardens lemma

State elimination method:-

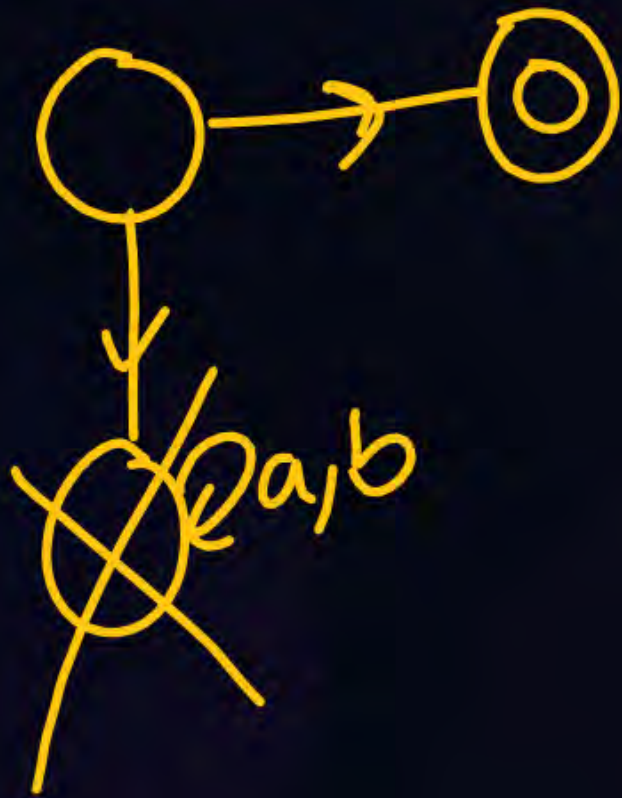
→ It work for E-NFA, NFA and DFA

Simplify the FA

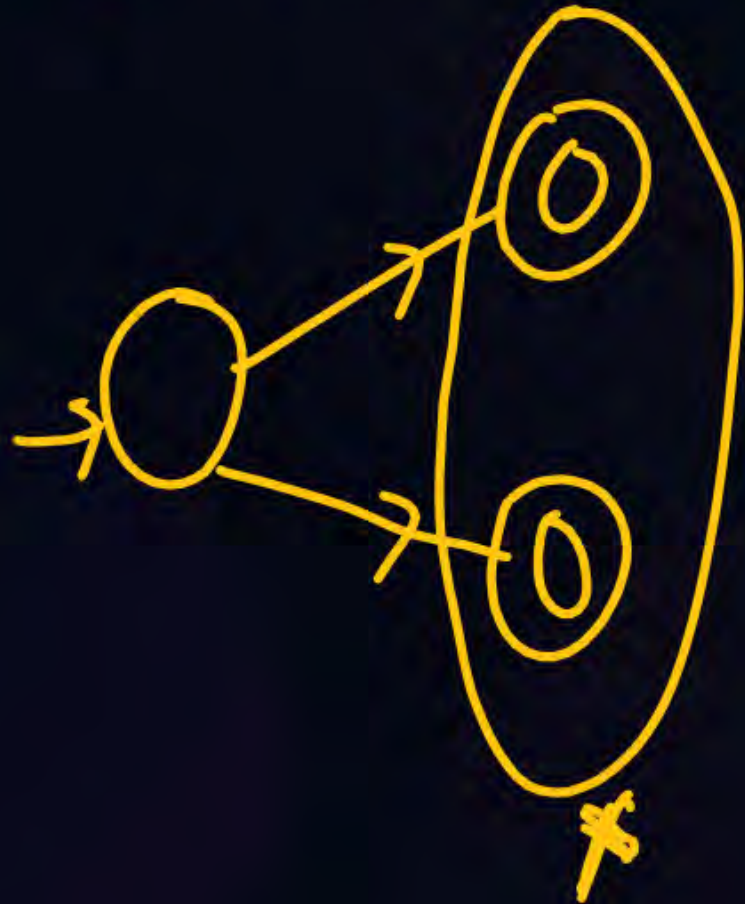
a) Remove non reachable states



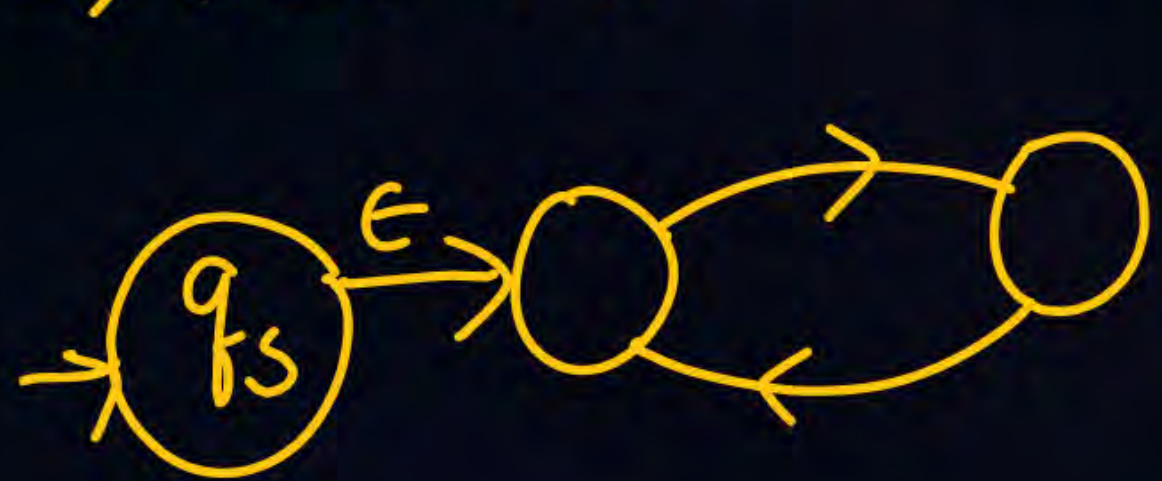
b) Remove dead states



c) merge final states if possible

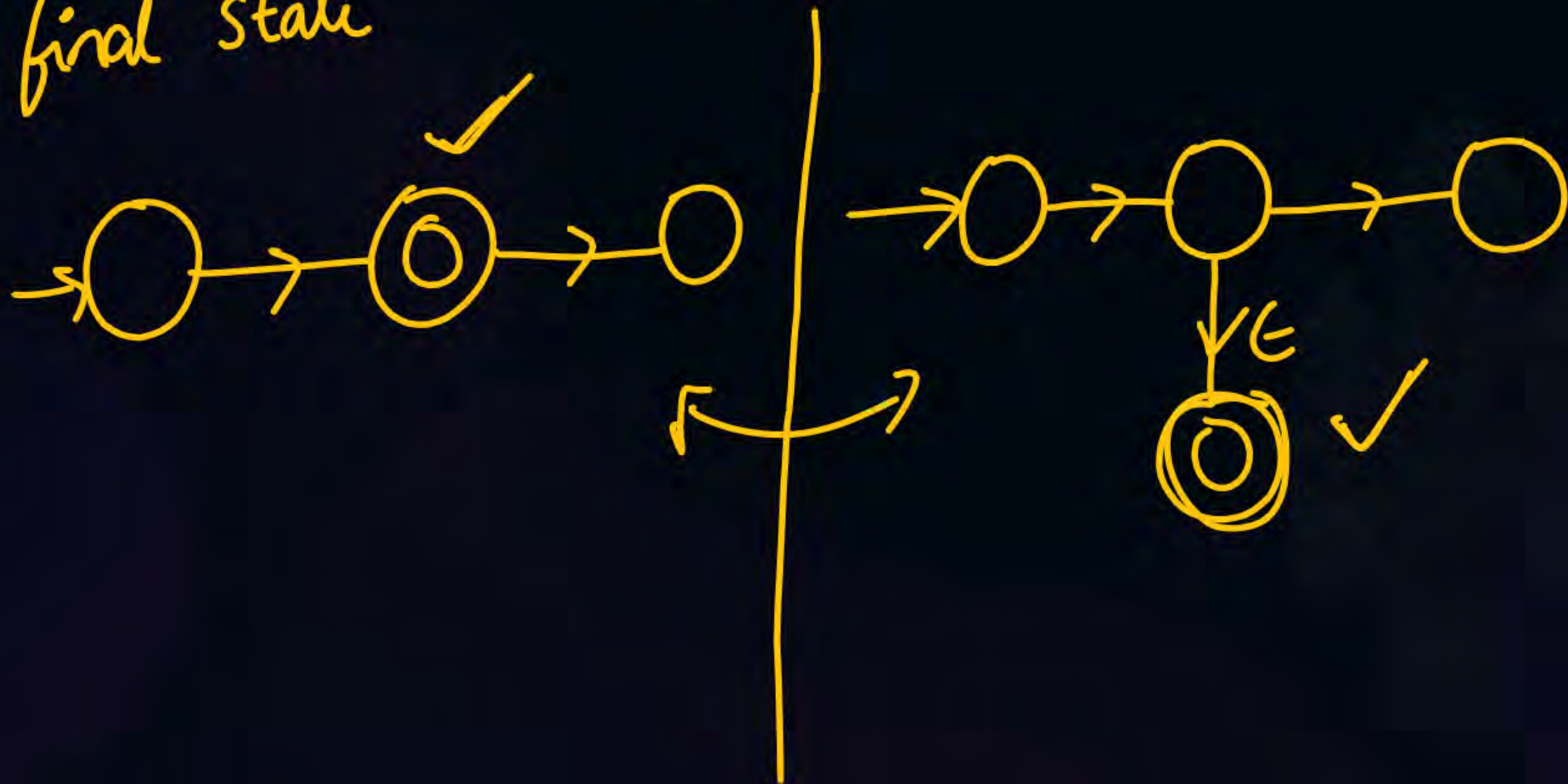


→ There should not be any incoming edges to initial state



Q → incoming
outgoing ✓

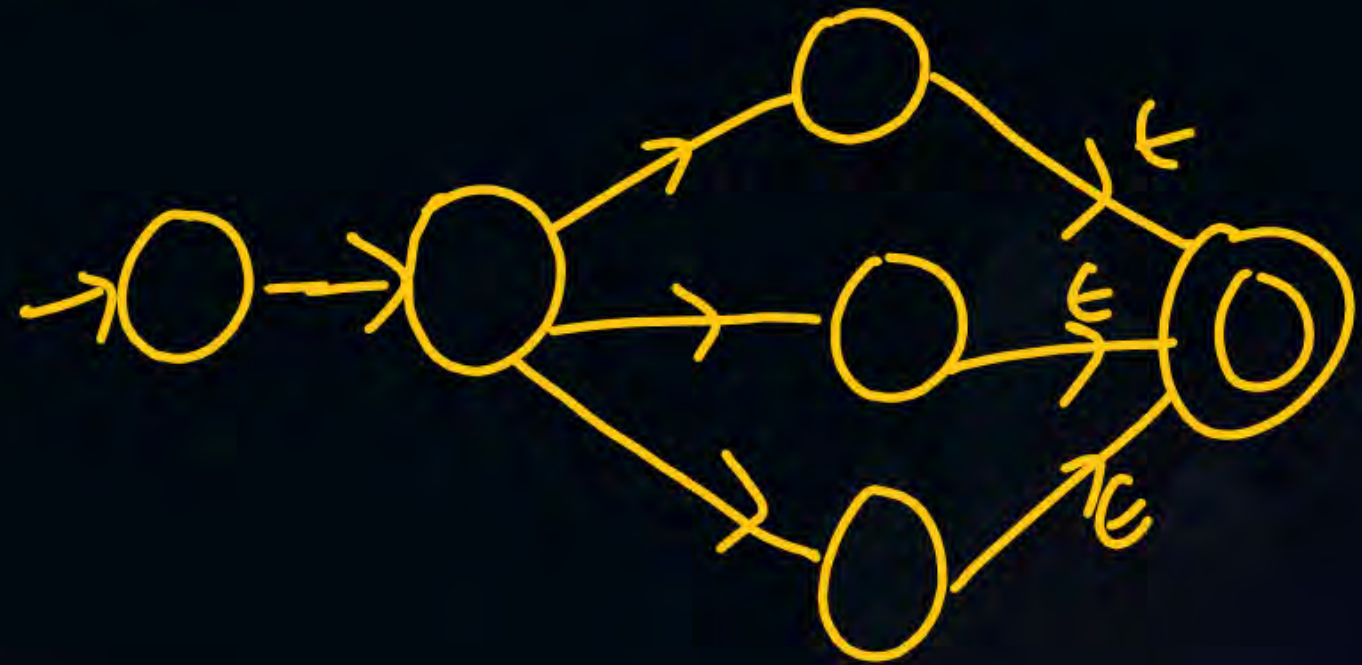
There should not be any outgoing edge from the final state



There should be only one final state

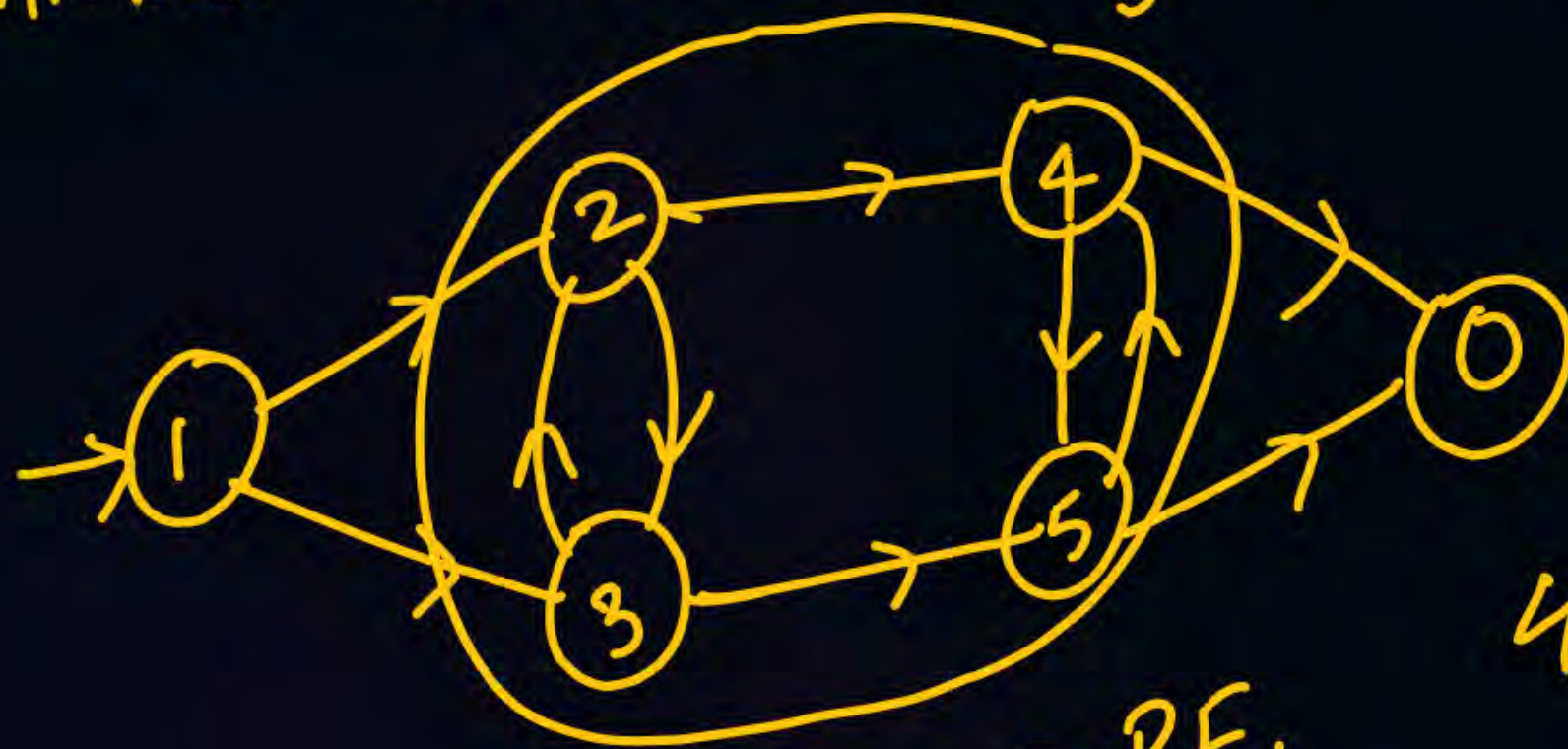


$\Rightarrow$



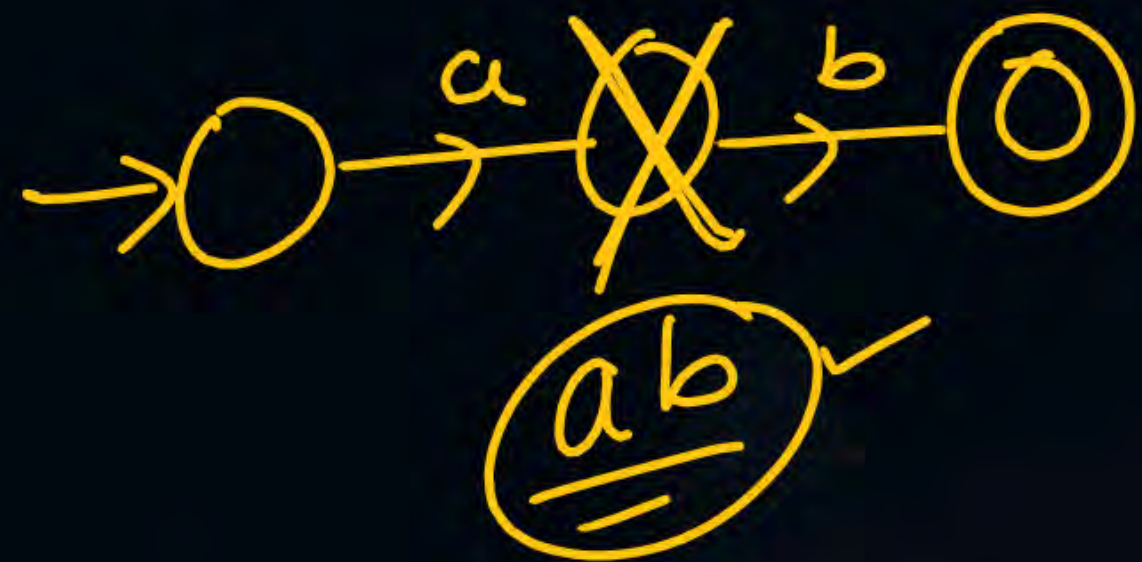
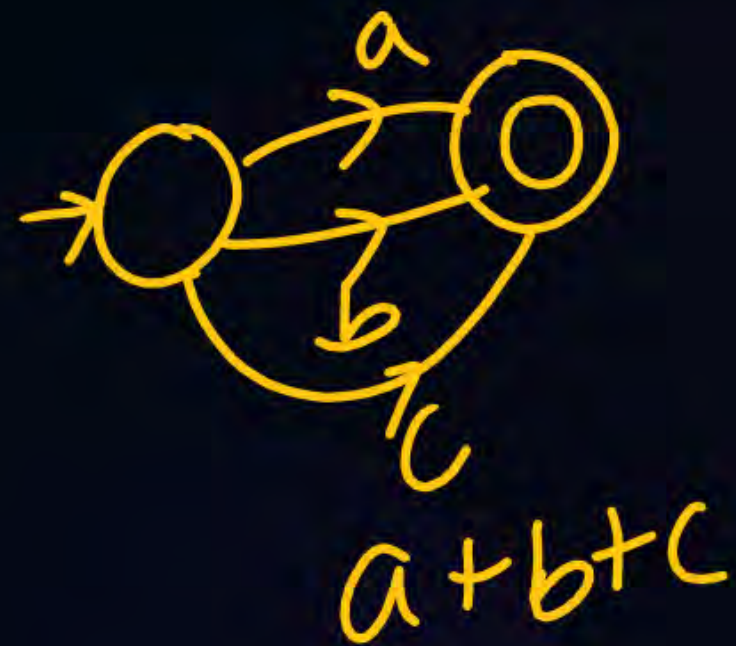
Big question
 FA $\rightarrow$ language $\rightarrow$ RE ✓

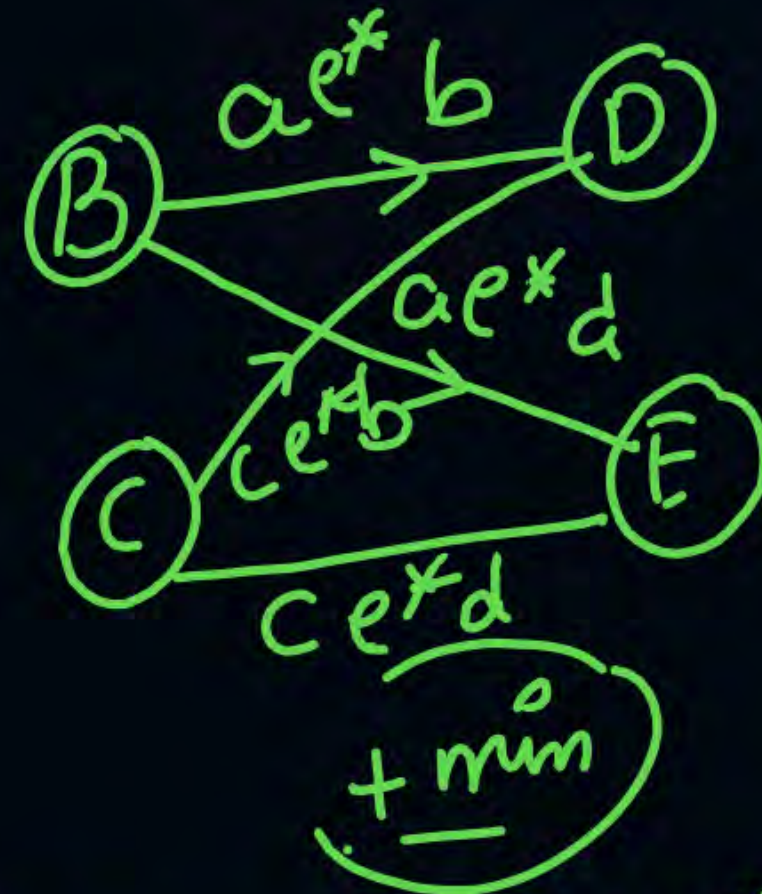
Eliminate states in any order, except initial state and final state



2 3 4 5 $\rightarrow$ RE₁
 3 2 4 5 $\rightarrow$ RE₂

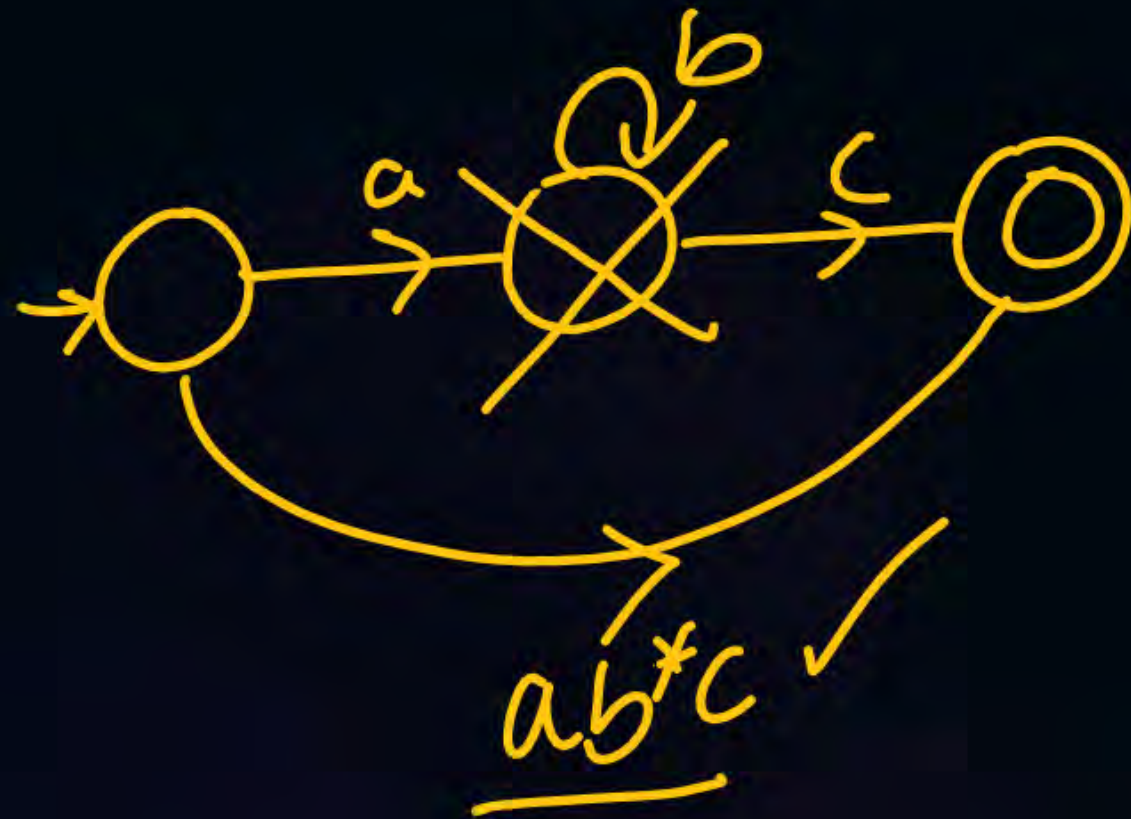
4 5 3 2 $\rightarrow$ RE₃ ✓

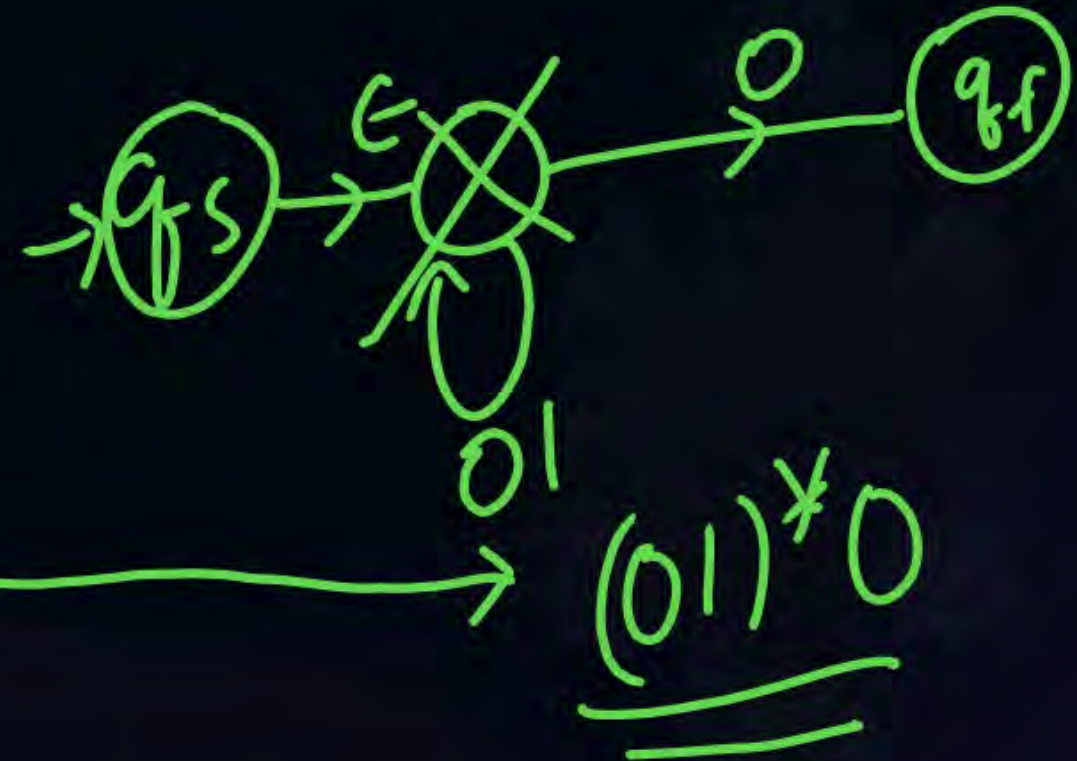
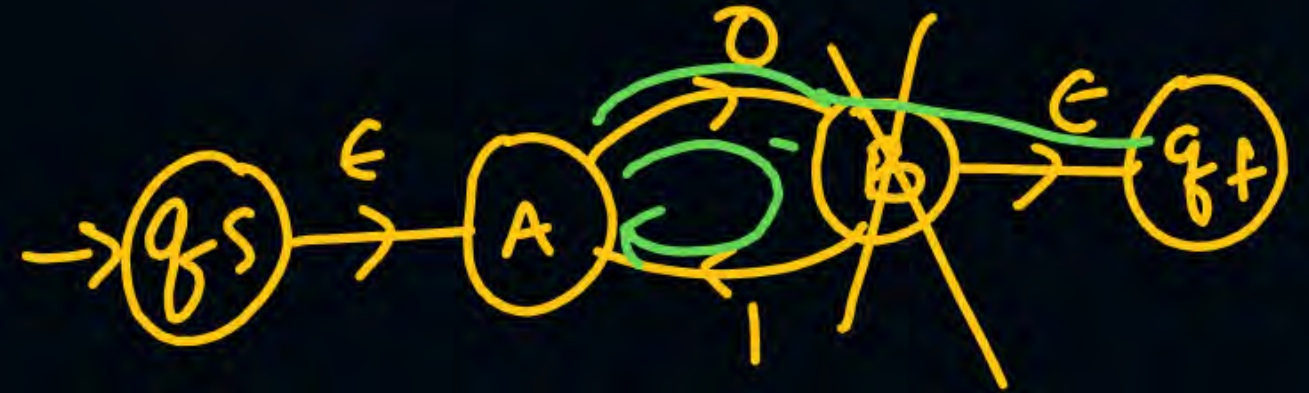




"A"

$A \rightarrow \text{In} \rightarrow (a, c)$
 $A \rightarrow \text{out} \rightarrow (b, d)$
 $(2 \times 2) \rightarrow \text{upath}$



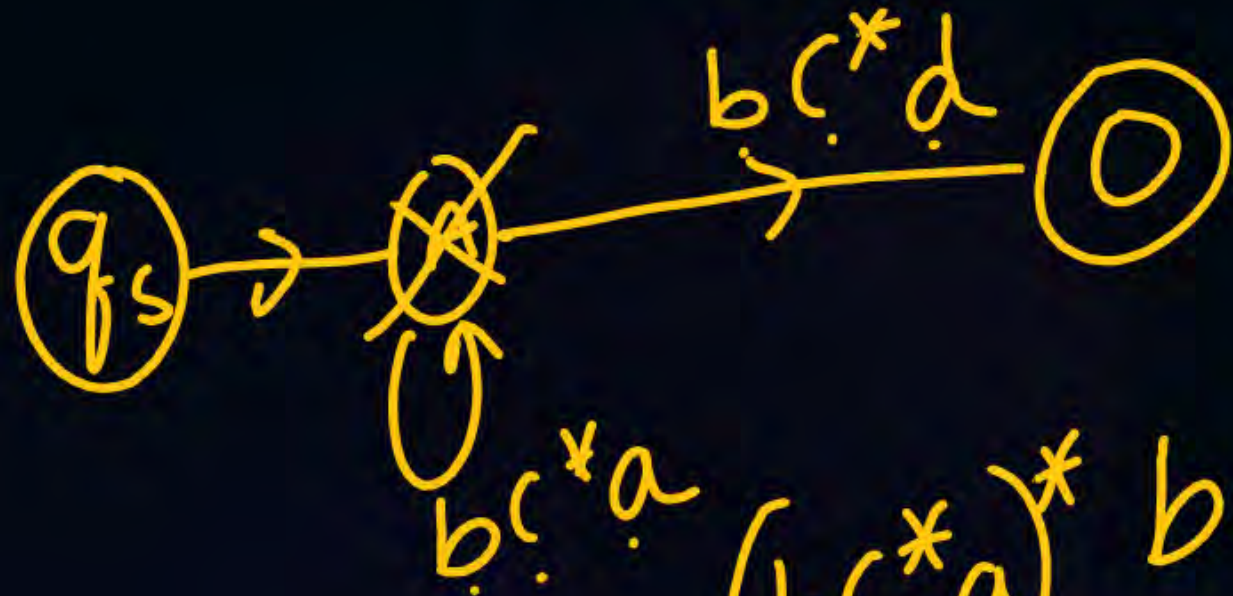


$0(10)^*$ ✓

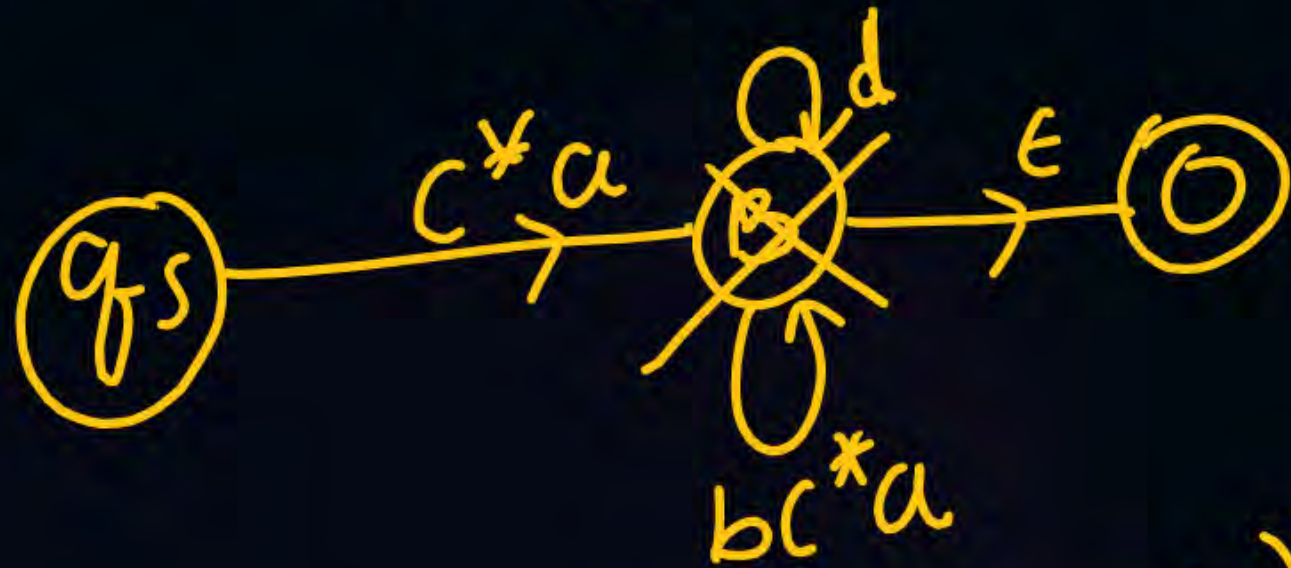
$(01)^*0$



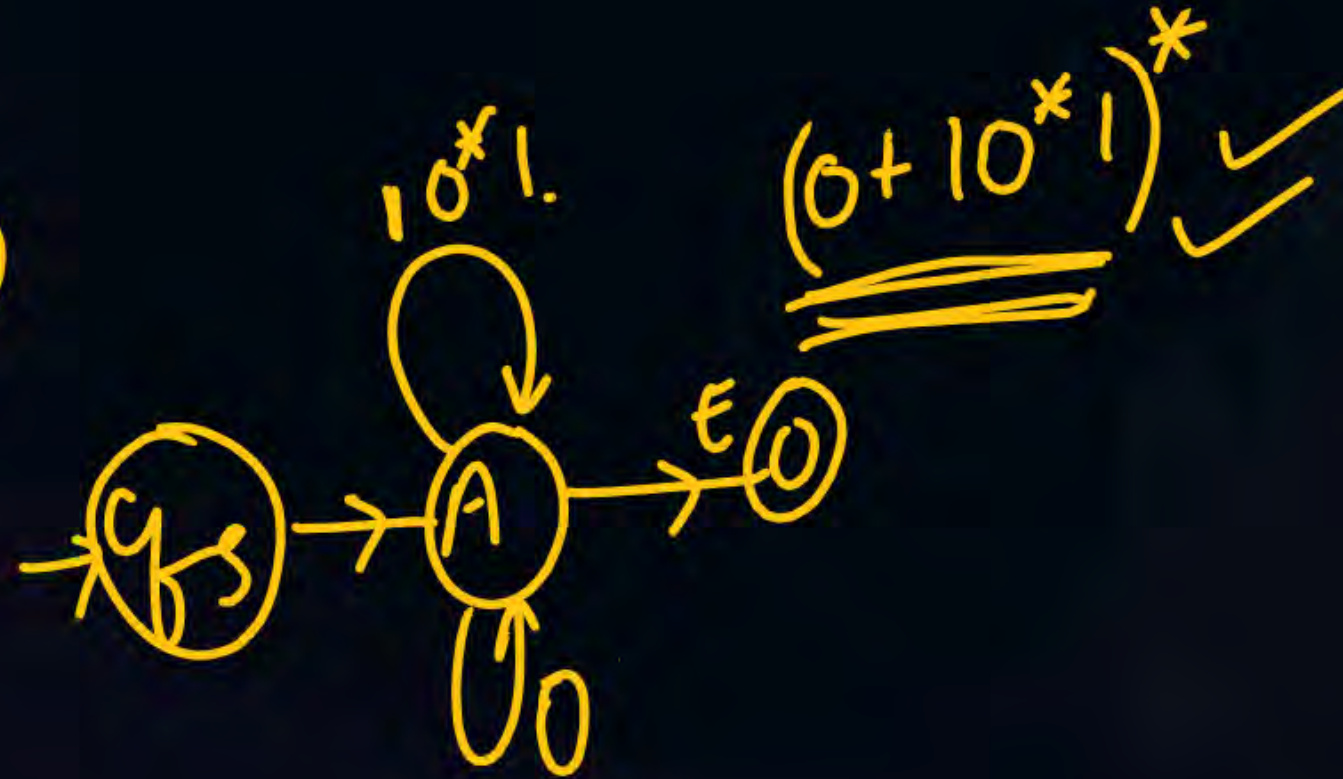
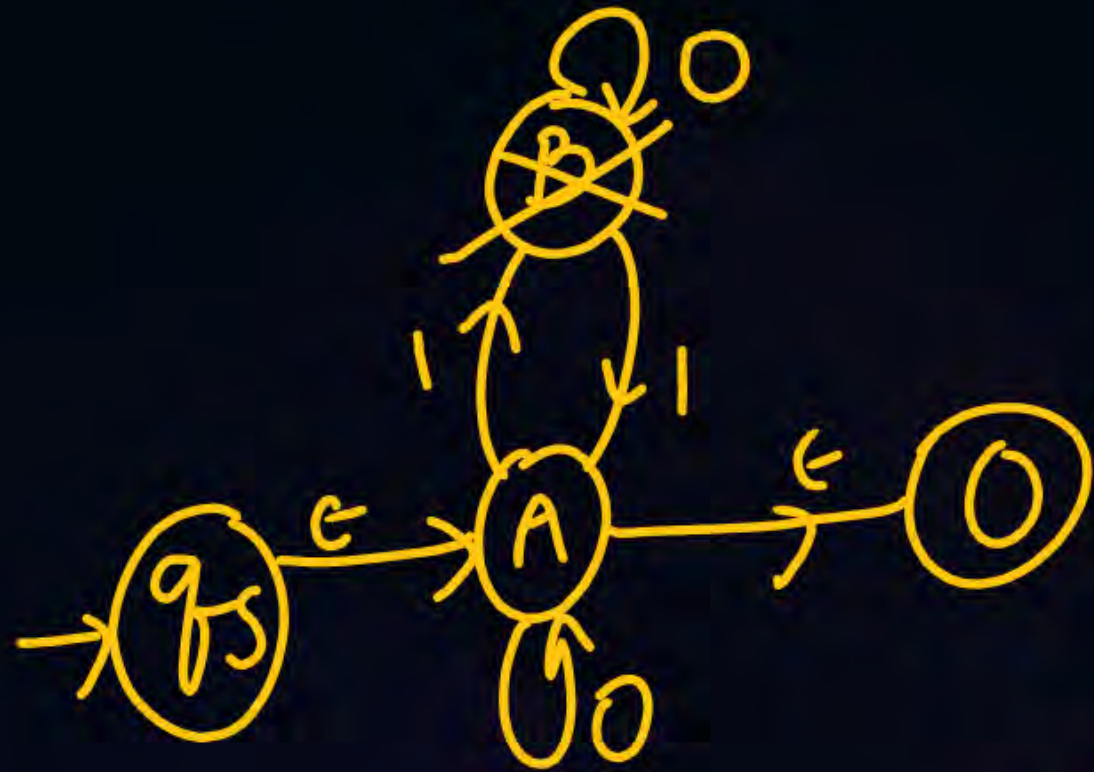
BA
AB ✓

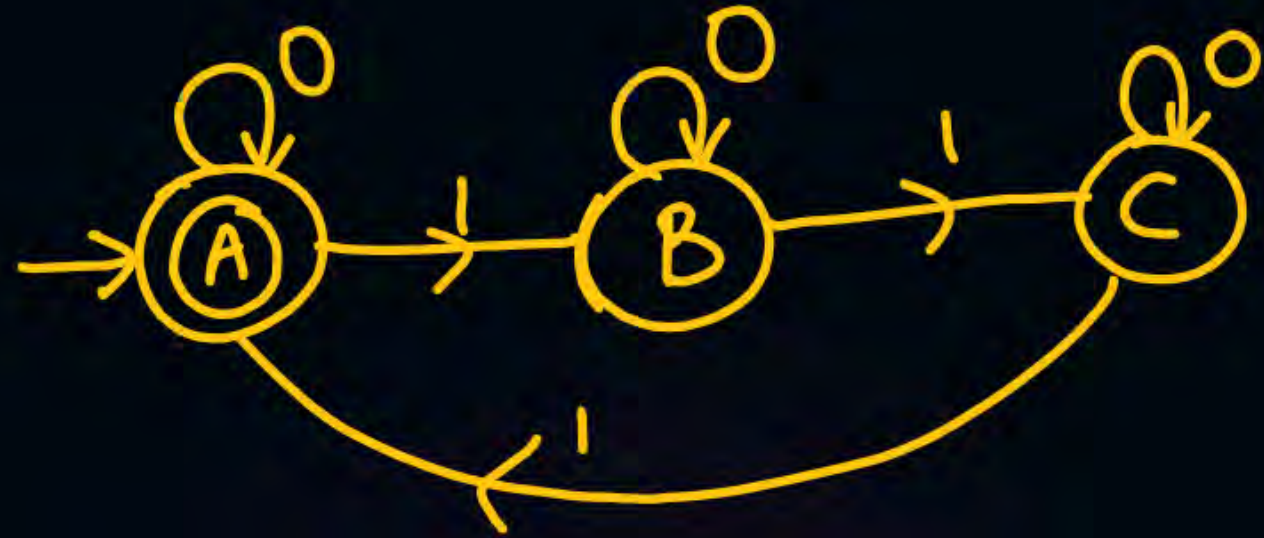


$(bc^*a)^*bc^*d$

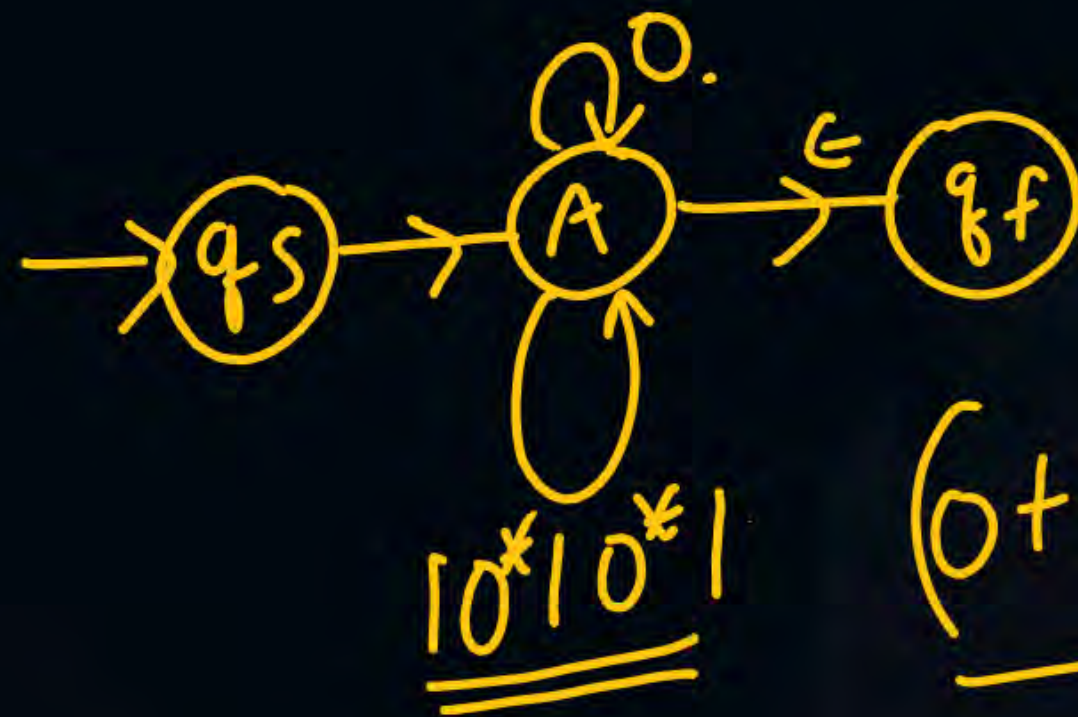
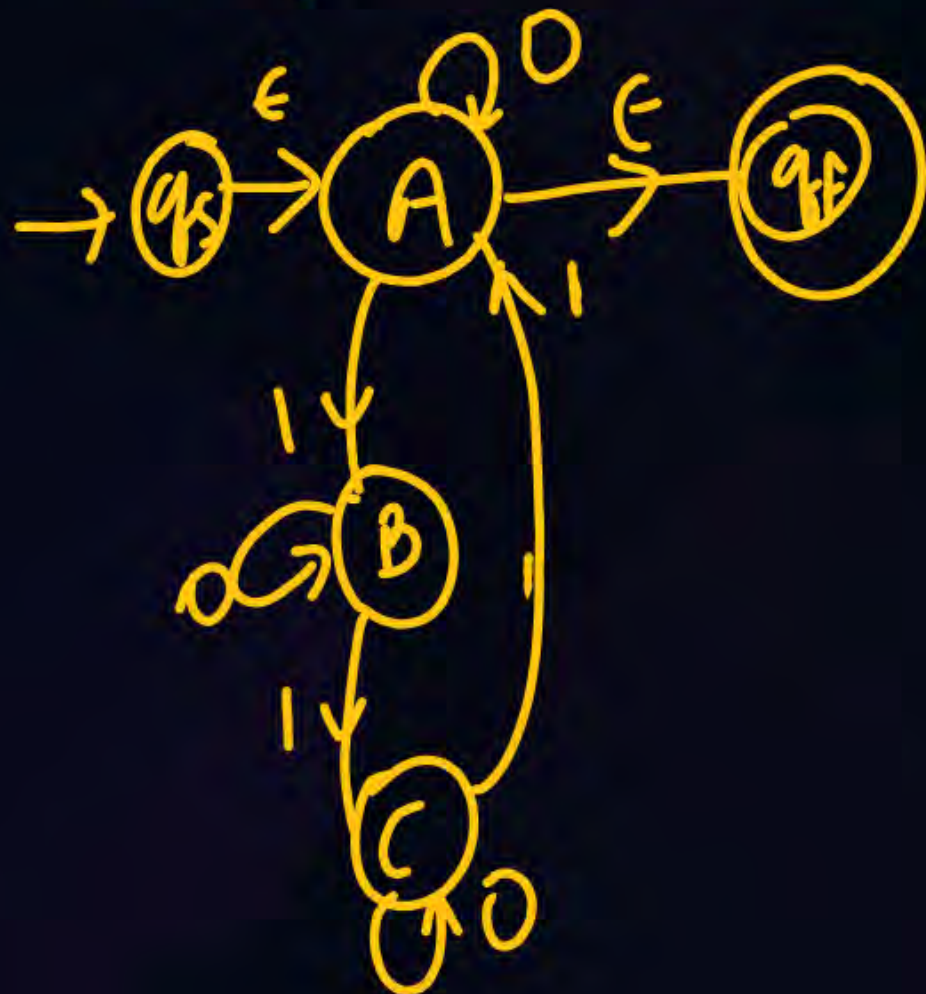


$$\underline{\underline{c^*a(d + b(c^*a))^*}}$$





→ 1's divisibly by "3"



$(0 + 10^*10^*1)^*$

Alders lemma :-

$$\textcircled{S = R + ST}$$

$$\textcircled{\underline{S = RT^*}}$$

$$RT^* = R + (RT^*)T$$

$$= R + RT^+$$

$$= R(E + T^+)$$

$$= RT^*$$

$$S = R + TS$$

$$\underline{S = T^*R}$$

$$T^*R = R + T(T^*R)$$

$$= R + T^+R$$

$$= (E + T^+)R$$

$$= \textcircled{T^*R} \checkmark$$



$$A = \epsilon$$

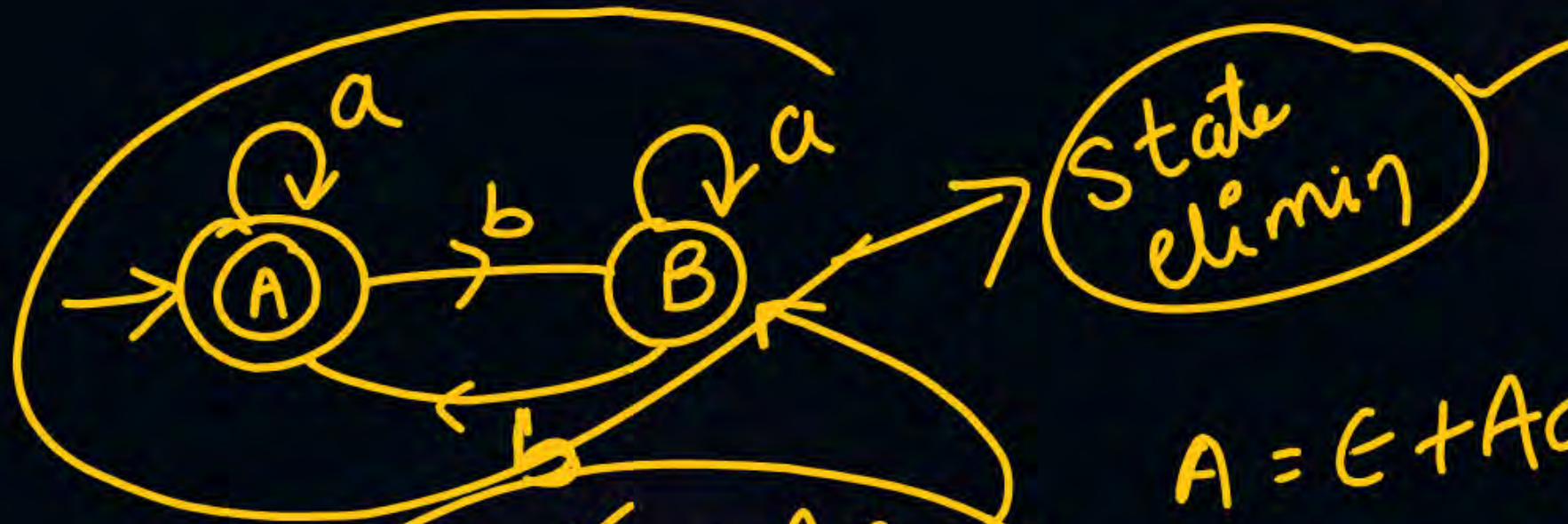
$$B = \textcircled{A}0 + B1 + C10$$

$$C = \textcircled{A}1 \checkmark$$

$$B = 0 + B1 + 10$$

$$B = \frac{(0+10) + B}{S \quad T}$$

$$S = RT^* \\ = \underline{(0+10)}1^* \checkmark$$



State
elimin

$$A = \epsilon + Bb + Aa$$

$$\frac{B}{S} = \frac{Ab}{R} + \frac{Ba}{ST}$$

$$S = RT^* = \underline{(Ab)a^*} = B$$

$$A = \epsilon + Aa + Aba^*b$$

$$\frac{A}{S} = \frac{\epsilon}{R} + \frac{A}{S} \frac{a + ba^*b}{T}$$

$$\underline{S = RT^*}$$

$$A = \epsilon (a + ba^*b)^* = \underline{(a + ba^*b)^*}$$

$$A = 10 + 01A + \underline{B} \quad (A \rightarrow \text{final})$$

$$B = 00 + CB$$

$$C = \frac{11}{R} + \frac{0C}{T \ S}$$

$$S = T * R$$

$$C = (0^*11)$$

=

$$\frac{B}{S} = \frac{00}{R} + \frac{0^*11B}{T \ S}$$

$$S = T * R$$

$$B = \frac{(0^*11)^*00}{\checkmark}$$

$$A = 10 + 01A + (0^*11)^*00$$

$$\frac{A}{S} = \frac{10 + (0^*11)^*00}{R} + \frac{01A}{T \ S}$$

$$S = T * R$$

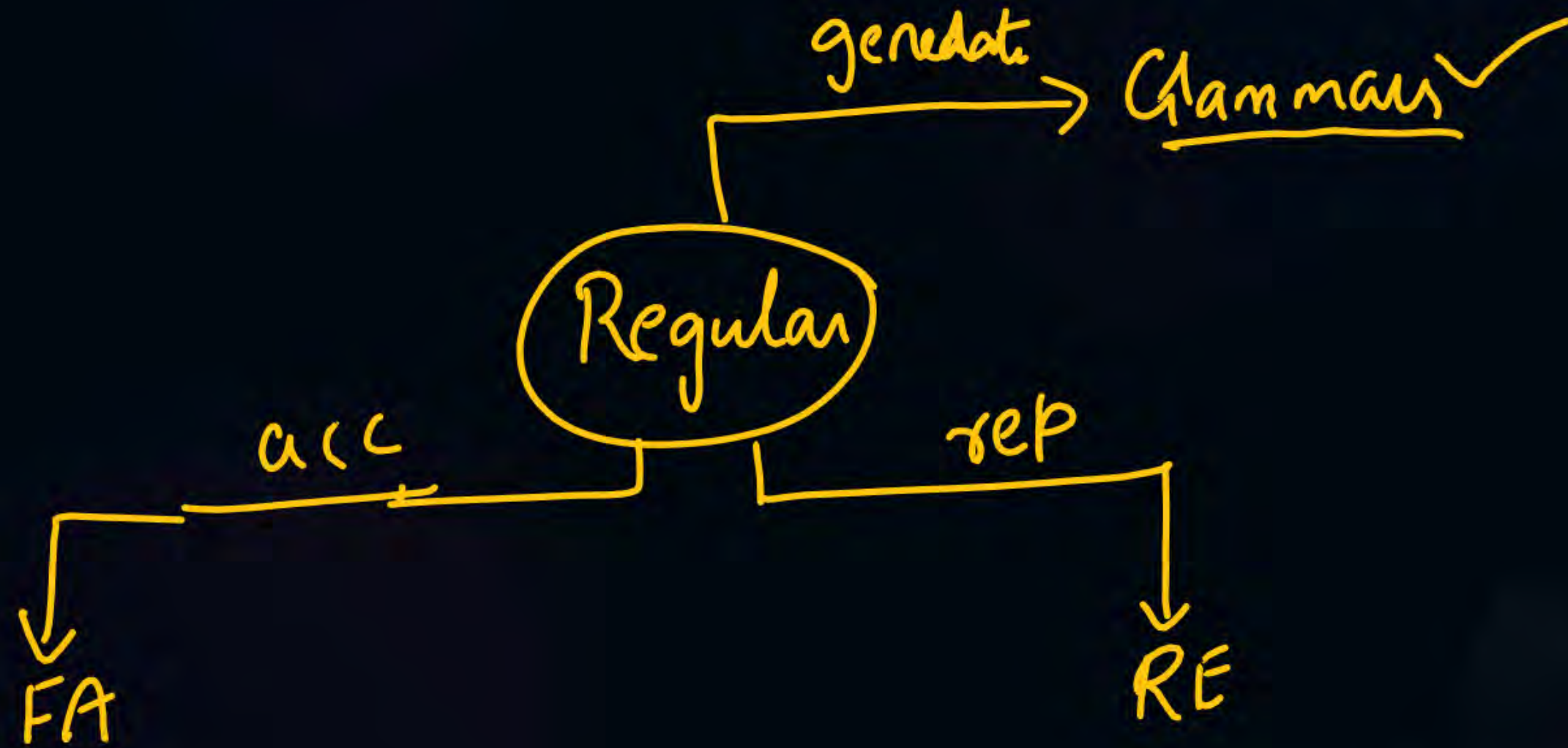
$$A = \underline{(01)^* (10 + (0^*11)^*)}$$



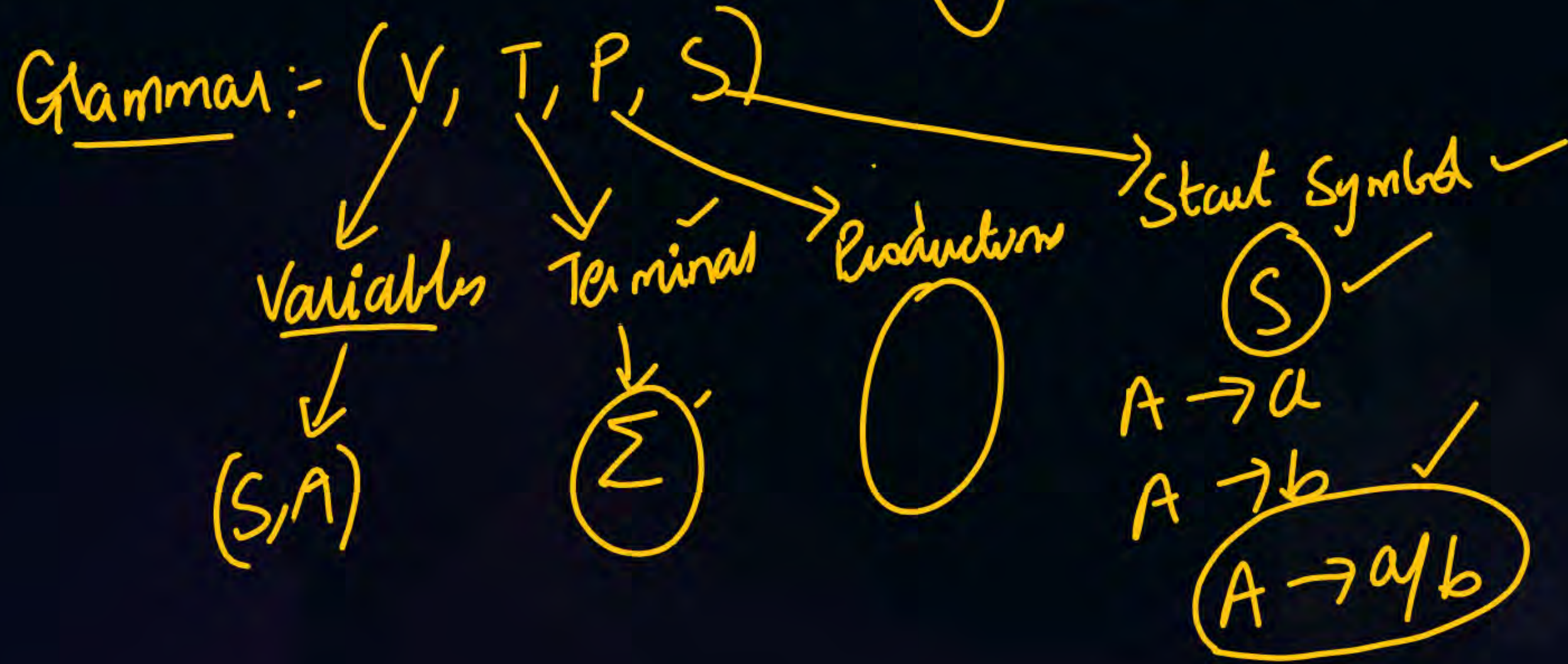
If D is final state, then language $\rightarrow L_D$
 If E is final state, then language $\rightarrow L_E$

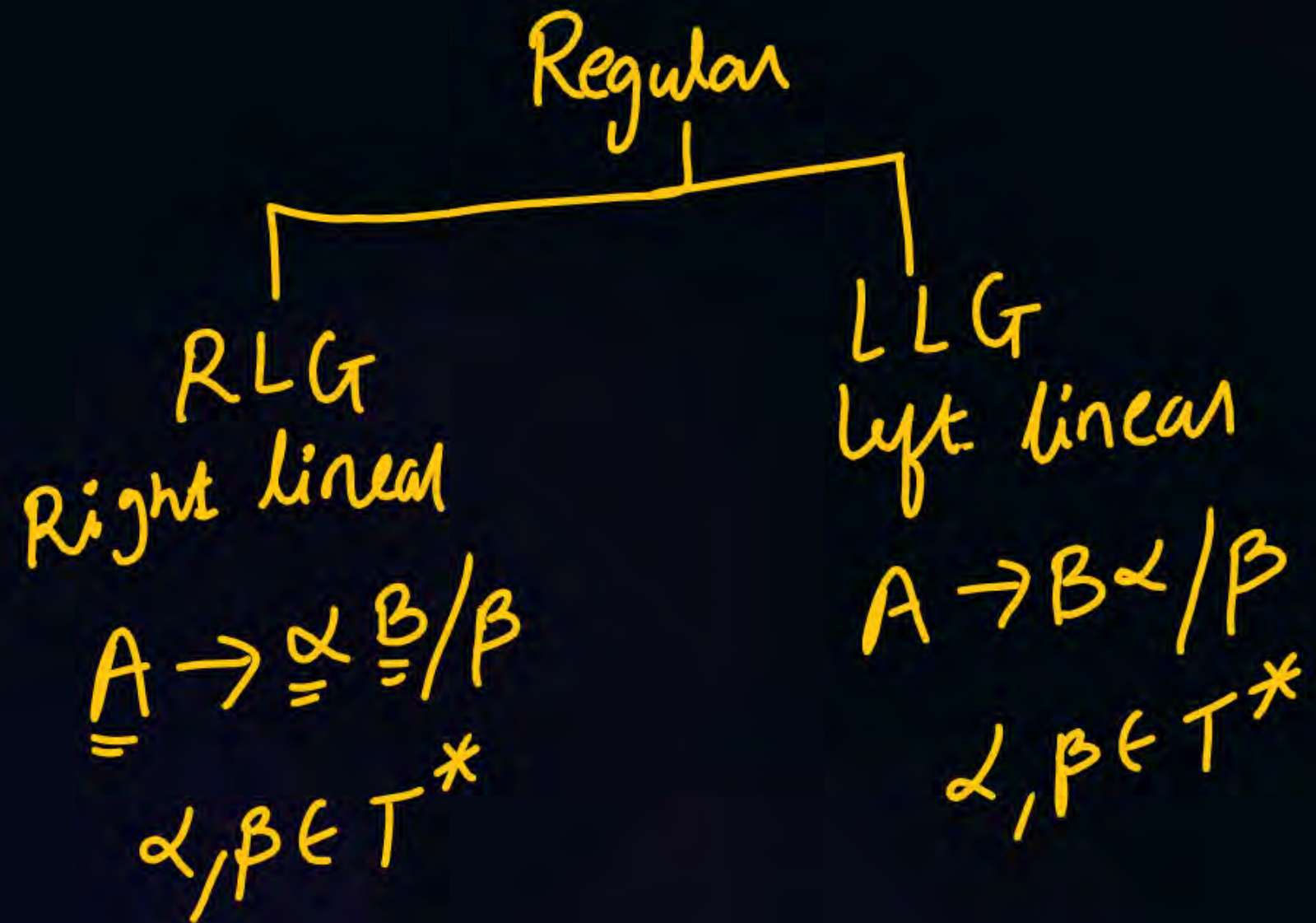
$$\begin{aligned}
 &\checkmark D = CO + DI + EO \\
 &\checkmark E = CO + EO + \overline{DI}
 \end{aligned}$$

- ~~a) $L_D = L_E$~~ c) $L_E \subseteq L_D$
 b) $L_D \subseteq L_E$ d) $L_D \neq L_E$



$$\begin{aligned} S &\rightarrow abS/aA \\ A &\rightarrow aSA/bA \end{aligned}$$





$$A \rightarrow a \underline{A} / b$$

RLG

$$B \rightarrow \underline{B} a / b$$

LLG

$$A \rightarrow a \underline{A} a / b \checkmark$$

Not regular

$$A \rightarrow a \underline{A} / \underline{B} b / b$$

Not reg x

$A \rightarrow aa\bar{A}/b$ ✓

RLG

$A \rightarrow \bar{A}aa/\bar{B}a/b$ ✓
 LLG

$S \rightarrow a\bar{S}b/ab \rightarrow$ CFG

Not RLG
 Not LLG

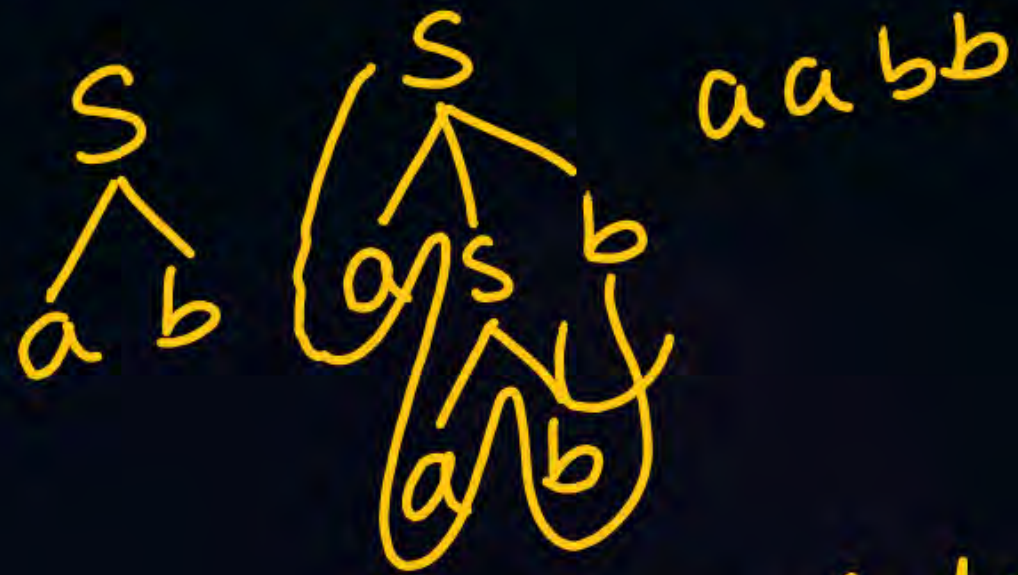
L_1 is reg $\begin{cases} \rightarrow \text{at least 1 RLG} \\ \rightarrow \text{at least 1 LLG} \end{cases}$

Every reg language will have at least one Regular gram

Some non regular grammars also generate
regular languages

$\begin{cases} \rightarrow \text{1 RLG} \\ \rightarrow \text{1 LLG} \end{cases}$

$S \rightarrow aSb / ab$



Derivation tree



aaaabbbb

$$S \rightarrow aSb / ab$$

Derivation

$$S \Rightarrow ab$$

$$S \Rightarrow a \underset{\uparrow}{S} b \Rightarrow aabb$$

$$S \Rightarrow a \underline{S} b \Rightarrow aa \underline{\underline{S}} bb \Rightarrow aaabbb$$

$A \rightarrow A/a$



Both RLK
LLK

$A \rightarrow a/b/c$



Both RLK
LLK

THANK - YOU